Vasomotor Symptoms of Menopause: A Comprehensive Disease Characteristics Report

Summary

Vasomotor symptoms of menopause (VMS) — hot flushes/flashes and night sweats — are an **acquired, non-Mendelian thermoregulatory disorder** of the menopausal transition. There is **no causal gene** and no OMIM/MONDO entry; the nearest coded anchor is **ICD-10-CM N95.1** (menopausal and female climacteric states). The initiating lesion is **withdrawal of ovarian estradiol** (CHEBI:16469 17β-estradiol). Estrogen loss de-represses a discrete population of hypothalamic arcuate-nucleus neurons co-expressing **kisspeptin, neurokinin B (NKB), and dynorphin** — the **KNDy neurons** — which become hypertrophic and hyperactive. These neurons signal via **neurokinin B acting on the NK3 receptor (NK3R), encoded by *TACR3***, onto thermoregulatory neurons in the hypothalamic preoptic area, narrowing the "thermoneutral zone" so that trivial elevations in core body temperature trigger inappropriate heat-dissipation responses — cutaneous vasodilation and sweating — perceived as a hot flash.

Although VMS has no Mendelian cause, human genetics, pathophysiology, and pharmacology converge on a single molecular node: a trans-ethnic GWAS identified ***TACR3*** (the NK3R gene) as the only genome-wide-significant susceptibility locus, and **NK3R antagonists (fezolinetant, and the dual NK1/NK3 antagonist elinzanetant)** are now proven non-hormonal therapies. This convergence makes VMS an unusually mechanistically "solved" common condition despite the absence of a disease gene.

VMS affect **50–80% of women**, persist for a **median of 7.4 years** (longest in Black/African American women, median 10.1 years), and are not merely a nuisance: frequent/persistent VMS mark an **adverse cardiovascular risk profile and elevated CVD/stroke event risk**, and cluster with **sleep disturbance and mood symptoms** into a triad that drives quality-of-life impairment. Treatment is led by **menopausal hormone therapy (MHT)** — most effective, favorable risk/benefit for women <60 years or within 10 years of menopause ("timing hypothesis") — with non-hormonal options including **NK3R antagonists** (~58% achieve ≥50%

VMS reduction), **SSRIs/SNRIs, gabapentin, and cognitive behavioral therapy (CBT)**. The NK3R antagonist class carries a **hepatic (transaminase-elevation) safety signal** requiring liver monitoring.

Key Findings

Finding 1 — Core mechanism: estrogen withdrawal → KNDy hyperactivity → NK3R signaling to the preoptic heat-defense pathway

The central pathophysiology of VMS is now well established from convergent human neuropathology, animal models, and reverse-pharmacology. Postmenopausal women show marked **hypertrophy and increased neurokinin B expression** in arcuate-nucleus KNDy neurons. When estradiol declines, these neurons — normally restrained by estrogen negative feedback — become hyperactive. They project from the arcuate nucleus to the **median preoptic nucleus** of the hypothalamic preoptic area, the brain's heat-defense integrator, where NKB acting on **NK3R** triggers heat-loss effectors: cutaneous vasodilation and sweating.

Rance et al. (PMID: 23872331) summarize the animal evidence: *"activation of NK_3R in the median preoptic nucleus, part of the heat-defense pathway, reduces body temperature. Finally, ablation of KNDy neurons reduces cutaneous vasodilatation and partially blocks the effects of estrogen on thermoregulation."* A 2025 review (PMID: 40330614) states the full chain: *"the kisspeptin-neurokinin B-dynorphin (KNDy) neuron complex … becomes hyperactive when estradiol levels decrease. KNDy neurons project from the arcuate nucleus to thermoregulatory areas within the hypothalamic preoptic area, where heat loss mechanisms are triggered, including cutaneous vasodilation and sweating."* The therapeutic corollary (PMID: 42045091) is that *"VMS pathogenesis centers on the hypothalamic KNDy neuron network and is primarily mediated by Neurokinin B acting on the NK3 receptor"* — validated by the efficacy of NK3R antagonism.

Ontology anchors: GO:0001659 (temperature homeostasis), GO:0007204 signaling / cutaneous vasodilation (heat loss); CL kisspeptin/KNDy neuron of the arcuate nucleus; UBERON:0001882 (arcuate nucleus of hypothalamus), UBERON:0002661 (preoptic area); CHEBI: neurokinin B, 17β-estradiol (CHEBI:16469).

Finding 2 — Epidemiology: high prevalence, long duration, racial/ethnic disparity

VMS are among the most common conditions of midlife women. In the SWAN cohort (Study of Women's Health Across the Nation; n=3302 enrolled, 1449 with frequent VMS), Avis et al. (PMID: 25686030) found *"The median total VMS duration was 7.4 years. Among 881 women who experienced an observable FMP [final menstrual period], the median post-FMP persistence was 4.5 years."* Duration varied by onset timing: women symptomatic while still pre/early-perimenopausal had the longest course (median >11.8 years), while postmenopausal-onset symptoms were shortest (median 3.4 years).

A striking racial/ethnic disparity exists: *"Compared with women of other racial/ethnic groups, African American women reported the longest total VMS duration (median, 10.1 years)."* Part of this disparity is attributable to social stressors: everyday discrimination is associated with VMS in Black women (PMID: 38595299) — *"Higher levels of discrimination were associated with concurrent reporting of any (odds ratio [OR], 1.57 [1.31-1.89]) and frequent (≥6 d) VMS (OR, 1.55 [1.21-1.99])."* Prevalence is 50–80% (PMID: 40330614), with substantial negative quality-of-life impact.

Finding 3 — Genetic susceptibility maps to *TACR3*, the same receptor targeted by effective drugs

While VMS is not a genetic disease, susceptibility has a heritable component mapping precisely to the drug target. A WHI trans-ethnic GWAS meta-analysis (n=17,695 European/African/Hispanic American women; Crandall et al. PMID: 28231077) found *"14 SNPs, all located on chromosome 4 in the tachykinin receptor 3 (TACR3) locus"* with $P < 5\times10^{-8}$, and *"These SNPs' effect sizes were similar across studies/participants' ancestry (odds ratio ~1.5). Genetic variation in TACR3 may contribute to the risk of VMS."* The SWAN Genomic Substudy (Zhao et al. PMID: 33906203) replicated and identified a protective allele: *"The C-allele of rs74827081 in TACR3 was associated with reduced likelihood of frequent VMS in White women (odds ratio [OR] = 0.49 [95% CI, 0.29-0.83])."* Polygenic risk scores for reproductive aging (age at menarche/menopause) also predicted VMS trajectories.

Gene annotation: *TACR3* (HGNC:11528), chromosome 4q24, encoding the NK3 receptor (NK3R), a class A GPCR; the natural ligand is neurokinin B (encoded by *TAC3*).

Finding 4 — Treatment efficacy: hormone therapy is most effective; NK3R antagonists are proven non-hormonal options

Menopausal hormone therapy (MHT) is the gold standard. Kaunitz & Manson (PMID: 26348174): *"Hormone therapy (HT) represents the most effective treatment for these symptoms with oral and transdermal estrogen formulations having comparable efficacy."* The **timing hypothesis** governs risk/benefit (PMID: 30484736): *"the risk of adverse cardiovascular events for HT are low for women <60 years of age or within 10 year from menopause."* Mehta (PMID: 33841322) concurs that risks are low for healthy women under 60 or within ten years of menopause.

Fezolinetant, a selective NK3R antagonist, is the first-in-class non-hormonal mechanistic therapy. In SKYLIGHT 2 (phase 3 RCT, NCT04003142; Johnson et al. PMID: 36734148): *"For VMS frequency, W4 least squares mean (SE) reduction vs placebo: fezolinetant 30 mg, -1.82 (0.46; P < .001); 45 mg, -2.55 (0.46; P < .001)"*, with benefit by week 1 maintained to week 52 and serious TEAEs ≤2%. Pooled SKYLIGHT 1+2 (PMID: 41259888): *"A greater proportion of fezolinetant-treated participants had a reduction of ≥50 % in VMS frequency from baseline to week 12 versus placebo (58.7 % versus 36.0 %, odds ratio [95 % CI]: 2.542 [1.868-3.472])."* In breast-cancer survivors where MHT is contraindicated, SSRIs/SNRIs and gabapentin are reasonable choices (PMID: 14663778).

Treatment	Class	Key efficacy evidence	PMID
Estrogen ± progestogen (MHT)	Hormonal	Most effective treatment; oral/transdermal comparable	26348174
Fezolinetant 45 mg	NK3R antagonist	≥50% VMS reduction in 58.7% vs 36.0% placebo (OR 2.54)	41259888, 36734148
Elinzanetant	Dual NK1/NK3 antagonist	~65% VMS reduction (OASIS program)	42045091
SSRIs/SNRIs (e.g., paroxetine)	Serotonergic/noradrenergic	Reasonable non-hormonal choice; MRS reduction vs placebo	14663778, 42171505
Gabapentin	α2δ calcium channel ligand	Reasonable non-hormonal choice	14663778
CBT	Behavioral	Hedges' g −0.60 to −0.87 on bother	42252432

Finding 5 — Phenotype: an episodic triad with sleep disturbance and mood symptoms

The VMS phenotype is **recurrent, episodic hot flashes** (sudden sensation of heat, flushing, sweating, sometimes palpitations/anxiety) and **night sweats**, each lasting ~1–5 minutes, often worst at night. VMS rarely occur in isolation. A real-world survey (Kingsberg et al. PMID: 38695491; 2451 physician forms, 1029 patient questionnaires) found that *"Significant burden from the triad of natural or induced menopausal symptoms, sleep disturbances and mood changes affected women's daily activities, work and quality of life more than vasomotor symptoms alone."* A treatment-preference study (PMID: 37650213) found *"86.5% and 87.8% reported moderate to very severe VMS and sleep problems during the preceding month,"* and that sleep improvement was the single most valued treatment attribute.

HPO suggestions: flushing (HP:0032324), abnormal/increased sweating — hyperhidrosis (HP:0000975), insomnia (HP:0100785), anxiety (HP:0000739), depressivity (HP:0000716), palpitations (HP:0001962).

Finding 6 — VMS as a cardiovascular risk biomarker

Frequent/persistent VMS are an observational biomarker of poorer cardiovascular health. Thurston (PMID: 37577812) summarizing SWAN + MsHeart/MsBrain: *"frequent and/or persistent VMS are associated with adverse CVD risk factor profiles, poorer underlying peripheral vascular and cerebrovascular health, and elevated risk for clinical CVD events."* Carson & Thurston (PMID: 37214424): *"VMS have been associated with multiple indicators of cardiovascular disease (CVD) risk, including an unfavorable CVD risk factor profile, increased subclinical CVD, and elevated risk of CVD events."* VMS are also linked to stroke risk (PMID: 42339545). These are prospective-cohort associations — **not proven causal**. Notably, in SWAN target-trial emulation (PMID: 42709460), MHT initiation in perimenopausal/recently postmenopausal women with VMS was associated with a 22% lower CVD risk (aHR 0.78, 95% CI 0.62–0.98), stronger within 10 years of menopause — though the authors caution against using this to justify MHT for CVD prevention.

Finding 7 — Animal model: the ovariectomized (OVX) rat with KNDy-neuron ablation

The standard induced model is the **ovariectomized rat** with **tail-skin-temperature (TSKIN)** readout as a hot-flash proxy. Mittelman-Smith et al. (PMID: 23150555) used the selective toxin **NK3-saporin** injected into the arcuate nucleus to ablate KNDy neurons: *"KNDy neuron ablation*

consistently reduced tail-skin temperature (T(SKIN)), indicating that KNDy neurons facilitate cutaneous vasodilatation, an important heat dissipation effector. Moreover, KNDy ablation blocked the reduction of T(SKIN) by 17β-estradiol (E(2))." This demonstrates KNDy neurons are **required** for cutaneous vasodilation and for estrogen's thermoregulatory effect — the causal spine of the human disease. **Species/orthologs:** rat (*Rattus norvegicus*, NCBI Taxon 10116) *Tacr3*; mouse (*Mus musculus*, 10090) *Tacr3*.

Finding 8 — Proximate physiology: a narrowed thermoneutral zone with elevated noradrenergic tone

At the whole-body level, Freedman (PMID: 24012626) frames the hot flash as *"a rapid and exaggerated heat dissipation response … triggered by small elevations in core body temperature (Tc) acting within a greatly reduced thermoneutral zone, i.e., the Tc region between the upper (sweating) and lower (shivering) thresholds. This is due in part, but not entirely, to estrogen depletion at menopause. Elevated central sympathetic activation, mediated through α2-adrenergic receptors, is one factor responsible for narrowing of the thermoneutral zone."* Crucially, absolute estradiol does not distinguish symptomatic from asymptomatic women (PMID: 15852197): *"the appearance of hot flashes coincides with estrogen withdrawal, this does not entirely explain the phenomenon because estrogen levels do not differ between symptomatic and asymptomatic women."* Central sensitivity — not estradiol level — determines who is symptomatic, explaining why *TACR3* genotype and noradrenergic tone matter.

Finding 9 — Non-hormonal/behavioral management: CBT and psychosocial interventions

Psychosocial interventions meaningfully reduce hot-flash bother. A meta-analysis (Robinson et al. PMID: 42252432; 28 RCTs, 2,887 women): *"Psychosocial interventions produced medium-to-large reductions in the bothersomeness of hot flushes and night sweats at short-term follow-up (Hedges' g = - 0.60 to - 0.87) and medium-term follow-up (g = - 0.50 to - 0.77)."* A CBT systematic review (PMID: 41462190; 16 studies, 910 women) found *"CBT significantly improves health-related quality of life and alleviates vasomotor, psychological, and sleep-related symptoms,"* with group-based CBT most effective. A CBT-for-menopausal-insomnia pilot RCT (PMID: 42084929): *"CBT-MI compared with MEC significantly reduced ISI (CBT-MI vs. MEC: -10.2±1.15 vs. -6.2±0.99; P=0.007), HFDIS (CBT-MI vs. MEC: -1.3±0.29 vs. -0.5±0.17; P=0.01)."* Physiology-based measures — ambient cooling, paced respiration, clonidine (α2-agonist) — also reduce flashes.

Finding 10 — Diagnosis is clinical; no confirmatory biomarker

VMS are **diagnosed clinically** from characteristic self-reported symptoms during the menopausal transition; **no laboratory or imaging test confirms VMS**. Objective ambulatory sternal skin-conductance monitoring quantifies flashes in research but is not routine. Hormonal testing has a narrow role: *"Measuring serum follicle stimulating hormone (FSH) and estradiol levels are helpful when in doubt"* (PMID: 24338340) to define menopausal status when the picture is ambiguous — not to diagnose VMS. Differential diagnosis of secondary flushing: thyroid dysfunction, carcinoid syndrome, pheochromocytoma, infection, anxiety/panic disorder, and medication effects.

Finding 11 — Etiology and risk factors: estrogen withdrawal is the cause; induced menopause is worst

The primary cause is **loss of ovarian estradiol** — natural, surgical, or iatrogenic. Induced menopause produces the most severe VMS: *"the unique situation of surgical menopause (with oophorectomy) and cancer therapy, conditions that are associated with frequent, severe, hot flashes"* (PMID: 40330614); *"Menopausal symptoms can be particularly severe due to the sudden loss of ovarian function"* (PMID: 34761721). Drivers of treatment-induced VMS (iVMS): bilateral oophorectomy, GnRH agonists, aromatase inhibitors, tamoxifen, and chemotherapy-induced ovarian failure. **Modifiable risk factors:** VMS severity associates with incident type 2 diabetes — *"incident T2DM was positively associated with increasing age, BMI, daily alcohol consumption, moderate-to-severe vasomotor symptoms (VMS, OR=1.045, 95% CI: 1.011 to 1.079)"* (PMID: 37671810); smoking and higher BMI increase risk; discrimination/psychosocial stress increase burden (PMID: 38595299). **Protective factors:** physical activity; the *TACR3* rs74827081 C-allele (genetic protective modifier).

Finding 12 — NK3R-antagonist safety: a hepatic signal requiring liver monitoring

Real-world pharmacovigilance has flagged a **hepatic (transaminase-elevation) signal** for the NK3R-antagonist class. A WHO VigiAccess analysis (Li et al. PMID: 42610452; 1,091 reports, 2,143 AEs): *"Liver-related abnormalities showed consistent signals, including increased alanine aminotransferase (ROR=20.60), aspartate aminotransferase (ROR=22.95), and bilirubin (ROR=12.52), suggesting a clustered hepatic signal."* A FAERS disproportionality analysis (PMID: 42547587) found serious liver injury as a regulatory concern across ROR/PRR/BCPNN/MGPS algorithms. An elinzanetant hepatic-safety review (Lewis et al. PMID: 41849128): *"The*

occurrence of elevated liver enzymes has been observed in clinical use with NK-3 receptor antagonists, e.g., fezolinetant and pavinetant, with recommendations for liver monitoring in the fezolinetant labeling information." In pivotal RCTs serious TEAEs remained infrequent (≤2%).

Mechanistic Model / Interpretation

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Ovarian follicular depletion / oophorectomy / endocrine cancer therapy → falling circulating estradiol (17β-estradiol)**. *(Demonstrated — the universal trigger.)*
2. Estradiol withdrawal **removes negative-feedback restraint on arcuate-nucleus KNDy neurons** → KNDy neurons **hypertrophy and overexpress neurokinin B** and become hyperactive. *(Demonstrated — human neuropathology,*

   P *23872331*

   .)
3. Hyperactive KNDy neurons **release excess neurokinin B** onto projections to the **median preoptic nucleus** of the hypothalamic preoptic area. *(Demonstrated in rodent tract-tracing/pharmacology.)*
4. Neurokinin B **activates NK3R (TACR3)** on preoptic thermoregulatory neurons → **destabilizes the heat-defense set-point and narrows the thermoneutral zone**, compounded by **elevated central α2-adrenergic/noradrenergic tone**. *(Demonstrated pharmacologically + physiologically,*

   P *24012626*

   .)
5. Within the narrowed thermoneutral zone, a **small rise in core body temperature crosses the sweating threshold → inappropriate heat-dissipation response**. *(Demonstrated.)*
6. Heat-dissipation effectors fire: **cutaneous vasodilation (flushing)** + **sweating** + **sometimes palpitations/anxiety** = the **hot flash / night sweat**. *(Demonstrated — effector physiology.)*
7. **Branch A (downstream sequelae):** nocturnal flashes → **sleep fragmentation → mood disturbance** → compounded QoL/work impairment (the triad).
8. **Branch B (associated risk marker):** frequent/persistent VMS **co-segregate with adverse vascular biology** → elevated CVD/stroke event risk *(association, causation unproven)*.

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Estradiol withdrawal (ovarian failure / oophorectomy / cancer Rx)
           │  removes estrogen restraint
           ▼
Arcuate KNDy neurons  ──hypertrophy, ↑ neurokinin B──►  HYPERACTIVE
           │  NKB release
           ▼
NK3R (TACR3) on median preoptic nucleus  ◄── ↑ central α2-adrenergic tone
           │  destabilized set-point
           ▼
NARROWED THERMONEUTRAL ZONE
           │  small ΔTc crosses sweating threshold
           ▼
Heat-dissipation response:  cutaneous vasodilation + sweating
           ▼
       HOT FLASH / NIGHT SWEAT
         ├──► sleep disruption ──► mood symptoms ──► ↓ QoL (triad)
         └──► marker of adverse vascular health ──► ↑ CVD/stroke risk (assoc.)

Therapeutic logic:
  • Estrogen (MHT) ........ restores step-2 restraint (most effective)
  • NK3R antagonist ....... blocks step-4 (fezolinetant / elinzanetant)
  • SSRI/SNRI, clonidine .. dampen noradrenergic component of step-4
  • CBT / cooling / paced respiration .. widen tolerance / raise threshold
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The elegance of VMS biology is that **three independent lines of evidence converge on NK3R/TACR3**: (a) human genetics (the sole GWAS locus is *TACR3*), (b) pathophysiology (KNDy→NK3R→preoptic area), and (c) pharmacology (NK3R antagonists work). This convergence is why VMS, despite being a common acquired condition with no disease gene, is mechanistically "solved" to the receptor level.

Section-by-Section Knowledge Base Content

1. Disease Information. Acquired thermoregulatory condition of the menopausal transition. Identifiers: no OMIM, no MONDO class, no Orphanet number; **ICD-10-CM N95.1** (menopausal and female climacteric states); **MeSH D019584 (Hot Flashes)**. Synonyms: VMS, hot flushes, hot flashes, night sweats, menopausal vasomotor symptoms. Information is drawn from **aggregated disease-level resources and cohort studies** (SWAN, WHI), not single-patient EHR.

2. Etiology. Cause: estradiol withdrawal. Genetic risk/modifier: *TACR3* (not causal). Environmental/lifestyle risk: induced/surgical menopause (strongest), smoking, obesity/high BMI, diabetes, alcohol, psychosocial stress/discrimination. Protective: physical activity, *TACR3* rs74827081 C-allele. Gene–environment interaction: central NK3R sensitivity (genotype) × degree/abruptness of estrogen loss (environment) determines symptom expression.

3. Phenotypes. Hot flashes (flushing HP:0032324), night sweats/hyperhidrosis (HP:0000975), palpitations (HP:0001962), insomnia (HP:0100785), anxiety (HP:0000739), depressivity (HP:0000716). Adult/perimenopausal onset; severity mild–severe (variable); course episodic/fluctuating; frequency 50–80% of women; median duration 7.4 y.

4. Genetic/Molecular. No causal gene. Susceptibility/modifier: *TACR3* (HGNC:11528, 4q24). Variants are common polymorphisms (e.g., rs74827081), not ACMG pathogenic variants; germline; population allele frequencies vary by ancestry with consistent ~1.5 OR. No chromosomal abnormalities. Epigenetics: estrogen-dependent regulation of HSP70/NO axis proposed
(P 28903474 ,
hypothesis-level).

5. Environmental. Iatrogenic triggers: oophorectomy, aromatase inhibitors, tamoxifen, GnRH agonists, chemotherapy. Lifestyle: smoking, obesity, alcohol, physical inactivity. No infectious agent.

6. Mechanism. See causal chain above. Molecular pathway: GPCR signaling (NK3R/Gq), noradrenergic/α2-adrenergic signaling. Cellular process: neuronal hyperexcitability (GO:0001659 temperature homeostasis). Cell type: KNDy/kisspeptin neuron (arcuate). No immune/metabolic primary defect (HSR/HSP70 link is hypothesis-level).

7. Anatomy. Primary: hypothalamus — **arcuate nucleus (UBERON:0001882)** and **preoptic area/median preoptic nucleus (UBERON:0002661)**. Effectors: **skin (UBERON:0002097)** cutaneous vasculature and **eccrine sweat glands (UBERON:0001820)**. Systems: nervous (CNS/autonomic), endocrine, integumentary, cardiovascular. Subcellular: plasma-membrane GPCR signaling (GO:0005886).

8. Temporal development. Onset: perimenopausal/adult, typically ages ~45–55; insidious. Course: episodic/fluctuating, chronic over years; median 7.4 y total, 4.5 y post-FMP; often remits spontaneously but may persist >10 y. Critical window: within 10 years of menopause (optimal MHT risk/benefit).

9. Inheritance and population. Not Mendelian — multifactorial/polygenic susceptibility. Prevalence 50–80% of menopausal women. Longest duration in African American women. Exclusively affects individuals undergoing menopause (overwhelmingly female; also relevant to androgen-deprivation/gender-affirming contexts).

10. Diagnostics. Clinical diagnosis; no confirmatory biomarker. FSH/estradiol only to clarify menopausal status when uncertain. Research tool: sternal skin-conductance monitoring. Rule out thyroid disease, carcinoid, pheochromocytoma, infection, anxiety/panic, drug effects.

11. Prognosis. Non-fatal, self-limited over years but with major QoL/sleep/mood morbidity; marker (not proven cause) of elevated CVD/stroke risk. Prognostic factors: onset timing (early perimenopausal onset → longer duration), race/ethnicity, BMI, smoking.

12. Treatment. MHT (most effective; NCIT: Estrogen Therapy / Hormone Replacement Therapy); NK3R antagonists fezolinetant and elinzanetant (dual NK1/NK3); SSRIs/SNRIs (paroxetine), gabapentin, oxybutynin, clonidine; CBT; cooling/paced respiration. Pharmacogenomics: *TACR3* genotype modulates baseline risk; fezolinetant metabolized to ES259564. Monitor liver enzymes on NK3R antagonists.

13. Prevention. No primary prevention of menopause itself. Where oophorectomy is elective, ovarian conservation / salpingectomy-with-ovarian-conservation avoids abrupt surgical VMS (P 34295957). P 34761721
Tertiary prevention: treat VMS to prevent sleep/mood/QoL sequelae; manage cardiometabolic risk factors. Behavioral: weight management, smoking cessation, physical activity.

14. Other species. Not a naturally occurring disease, but the **ovariectomized rat (NCBI Taxon 10116)** and mouse (10090) with tail-skin-temperature readout model the thermoregulatory physiology; orthologs *Tacr3*, *Tac3*, *Kiss1*. No veterinary natural-disease counterpart; no zoonosis.

15. Model organisms. Primary induced model: **OVX rat ± NK3-saporin KNDy ablation** (P 23150555) — recapitulates estrogen-dependent cutaneous vasodilation and KNDy requirement; limitation: rodents lack subjective hot-flash report; TSKIN is a proxy. Mouse *Tacr3*/*Kiss1* genetic models inform receptor biology.

Evidence Base

PMID	Study / type	How it supports the findings
23872331	Rance — mechanism review	KNDy→NK3R→preoptic heat-defense; KNDy ablation reduces vasodilation
40330614	2025 thermoregulation review	Full causal chain; prevalence 50–80%; induced menopause worst
42045091	Elinzanetant review	NKB–NK3R as primary mediator
25686030	SWAN natural history	Median duration 7.4 y; African American longest (10.1 y)
38595299	SWAN	Discrimination → VMS in Black women (OR ~1.55–1.57)
28231077	WHI trans-ethnic GWAS	*TACR3* sole genome-wide locus (OR ~1.5)
33906203	SWAN genomics	*TACR3* rs74827081 protective allele (OR 0.49)
36734148	SKYLIGHT 2 RCT	Fezolinetant efficacy, LS-mean −2.55 at W4
41259888	Pooled SKYLIGHT 1+2	≥50% responders 58.7% vs 36.0% (OR 2.54)
26348174	Review	MHT most effective
30484736	Review	Timing hypothesis
33841322	Review	HT risks low for women <60 / within 10 y
14663778	Review	SSRIs/gabapentin for breast-cancer survivors
38695491	Real-world survey	VMS–sleep–mood triad worse than VMS alone
37650213	Preference study	86.5% severe VMS, 87.8% sleep problems
37577812	SWAN/MsHeart review	VMS ↔ CVD risk/events
37214424	Review	VMS ↔ subclinical + clinical CVD
42709460	SWAN target-trial emulation	MHT in VMS women aHR 0.78 for CVD (timing-dependent)
42339545	Review	VMS ↔ stroke
23150555	OVX rat NK3-saporin	KNDy required for cutaneous vasodilation + estrogen effect
24012626	Freedman mechanism	Narrowed thermoneutral zone; α2-adrenergic

PMID	Study / type	How it supports the findings
15852197	Freedman	Estradiol level equal in symptomatic/asymptomatic
42252432	Meta-analysis	Psychosocial g −0.60 to −0.87
41462190	CBT review	CBT improves QoL/VMS/sleep
42084929	CBT-MI RCT	↓ ISI, ↓ hot-flash interference
24338340	Clinical	FSH/estradiol only when in doubt
37671810	Cohort	VMS severity ↔ T2DM (OR 1.045), BMI, alcohol
34761721	Toolkit	Surgical menopause → severe symptoms
42610452	WHO VigiAccess	Hepatic signal (ALT ROR 20.6, AST 22.95)
42547587	FAERS	Serious liver injury signal
41849128	Elinzanetant safety	NK3R class hepatic signal; liver monitoring

Limitations and Knowledge Gaps

- **No disease ontology anchor.** VMS lacks OMIM/MONDO/Orphanet entries; knowledge-base ingestion must use ICD-10-CM N95.1 and MeSH D019584 as proxies, which blur VMS with broader "climacteric states."
- **Causation vs. association for CVD.** The VMS↔cardiovascular/stroke link is consistently observed in prospective cohorts but **not established as causal**; shared upstream vascular/autonomic biology may drive both. The SWAN MHT target-trial emulation suggests benefit but with explicit caution about residual confounding.
- **Central sensitivity unexplained at the molecular level.** Because estradiol levels do not differ between symptomatic and asymptomatic women, the determinants of individual NK3R/noradrenergic set-point sensitivity (beyond *TACR3* genotype) remain incompletely defined.
- **Animal-model proxy limits.** Rodent tail-skin-temperature is an objective effector readout but cannot capture the subjective hot-flash percept or the night-sweat/sleep/mood sequelae.
- **NK3R-antagonist hepatotoxicity.** The hepatic signal is derived largely from disproportionality analyses (FAERS/VigiAccess), which establish signals but not incidence or

causation; long-term and oncology-population safety data for fezolinetant/elinzanetant are still maturing.

- **Underrepresentation and disparity drivers.** Mechanistic understanding of why African American women have longer, more severe VMS (beyond discrimination/psychosocial mediation) is incomplete.
- **Epigenetics/metabolic hypotheses unconfirmed.** The NO–HSP70/heat-shock-response hypothesis
(P 28903474)
linking estrogen deficiency to inflammatory disease is intriguing but not mechanistically proven in humans.

Proposed Follow-up Experiments / Actions

1. **Establish causality of the VMS–CVD link** via Mendelian randomization using *TACR3* and reproductive-aging polygenic scores as instruments, and via SWAN target-trial emulations — to determine whether treating VMS (e.g., with NK3R antagonists) alters cardiovascular endpoints.
2. **Prospective hepatic-safety registry** for fezolinetant/elinzanetant with scheduled ALT/AST/bilirubin monitoring to convert pharmacovigilance signals into incidence estimates and identify pharmacogenomic (CYP, *TACR3*) predictors of transaminitis.
3. **Genotype-stratified NK3R-antagonist trials** testing whether *TACR3* risk/protective alleles predict response magnitude — a precision-medicine test of the genetics↔pharmacology convergence.
4. **Dedicated trials in induced/surgical menopause and breast-cancer survivors** (where VMS is most severe and MHT contraindicated), with NK3R antagonists and oncologic safety endpoints.
5. **Mechanistic single-cell/spatial transcriptomics** of human hypothalamic arcuate and preoptic tissue across menopausal status to map KNDy hypertrophy and NK3R expression at cell-type resolution (CELLxGENE / Human Cell Atlas integration).
6. **Head-to-head and combination studies** of CBT + NK3R antagonist vs. either alone, targeting the full VMS–sleep–mood triad rather than hot-flash count alone.

7. **Disparities research** dissecting biological vs. social mediators of the longer VMS course in African American women, integrating discrimination measures with vascular and neuroendocrine phenotyping.

Report compiled from 5 iterations, 12 confirmed findings, and 44 reviewed papers. Evidence types span human cohort studies (SWAN, WHI), randomized controlled trials (SKYLIGHT, OASIS programs), human neuropathology, rodent models (OVX rat), and real-world pharmacovigilance.
